Diagonal maximality of the inverse geometric Vandermonde matrix

Principia Math in Progress

12 September 2026

Task 1. Solve the inverse Vandermonde conjecture. For $b > 1$ and $n \geq 1$, put

$$V = (b^{ij})_{0 \leq i,j < n}, \qquad C = V^{-1} = (c_{ij}), \qquad M_b(n) = \max_{i,j} |c_{ij}|, \qquad n_0 = \left\lceil \log_b(1 + b^{-1}) \right\rceil.$$

Sanna, Shallit, and Zhang conjectured that, for all sufficiently large n, $M_b(n) = |c_{ii}|$ for some $0 \leq i \leq n_0$ [1, Conjecture 11]. The accepted manuscript retains this as Conjecture 12. We prove the assertion for every n.

Theorem 1. *For every $b > 1$ and $n \geq 1$,*

$$M_b(n) = \max_{0 \leq i \leq \min(n-1, n_0)} c_{ii}.$$

All diagonal entries are positive, and no off-diagonal entry attains $M_b(n)$.

Proof. Write $s = \log b > 0$. For $x = (x_0, \ldots, x_{n-1}) \in \mathbb{R}^n$, the absolutely convergent exponential series gives

$$x^T V x = \sum_{k=0}^{\infty} \frac{s^k}{k!} \left(\sum_{i=0}^{n-1} x_i i^k \right)^2, \qquad (0^0 = 1).$$

If $x \neq 0$, at least one inner sum with $0 \leq k < n$ is nonzero: the Vandermonde matrix on the distinct nodes $0, 1, \ldots, n-1$ is invertible. Thus V, and hence C, is positive definite. In particular $c_{ii} > 0$, and each two-by-two principal minor yields

$$|c_{ij}|^2 < c_{ii} c_{jj} \quad (i \neq j). \tag{1}$$

Consequently $M_b(n) = \max_i c_{ii}$, and off-diagonal entries are strictly smaller than that maximum.

For completeness we verify the needed index bound, also obtainable from [1, Theorem 1]. Let e_m denote the elementary symmetric polynomial of degree m, with $e_0 = 1$, and set

$$E_m(j) = e_m(\{b^h : 0 \leq h < n,\ h \neq j\}), \qquad \pi_j = \prod_{\substack{0 \leq h < n \\ h \neq j}} |b^j - b^h|.$$

The coefficients of the Lagrange polynomial $\prod_{h \neq j} (X - b^h)/(b^j - b^h)$ form column j of C. Thus

$$|c_{ij}| = \frac{E_{n-1-i}(j)}{\pi_j}. \tag{2}$$

For $0 \leq j < n-1$, let $T = \{b^h : h \neq j, j+1\}$, and use $e_m(T) = 0$ outside its natural degree range. Then

$$E_m(j) - E_m(j+1) = (b^{j+1} - b^j) e_{m-1}(T) \geq 0. \tag{3}$$

Direct cancellation in the two products for π_j gives

$$\frac{\pi_{j+1}}{\pi_j} = \frac{b^{n+j-1} - b^{n-2}}{b^{n-1} - b^j} > 1 \quad \text{if } j \geq n_0. \tag{4}$$

Indeed the last inequality is equivalent to

$$b^j > \frac{b^{n-1} + b^{n-2}}{b^{n-1} + 1},$$

whose right side is strictly less than $1+b^{-1} \le b^j$. All numerators in (2) are positive. Equations (2)–(4) therefore show

$$|c_{i,j+1}| < |c_{ij}| \quad (0 \le i < n,\ n_0 \le j < n-1).$$

Using symmetry, for $n_0 \le j < n-1$ we obtain

$$c_{j+1,j+1} < |c_{j+1,j}| = |c_{j,j+1}| < c_{jj}.$$

Hence the diagonal strictly decreases after index n_0. If $n-1 \le n_0$, the index restriction is automatic; otherwise every larger diagonal entry is bounded by $c_{n_0 n_0}$. Together with (1), this proves the theorem. □

Task 2. Locate the maximizing diagonal index. Determine, as functions of b and n, which indices attain $\max_i c_{ii}$, and characterize the values of b at which ties occur.

Task 3. Quantify convergence to the limiting maximum. Find effective error bounds for $M_b(n)$ relative to its known limit, with explicit dependence on b, especially as $b \downarrow 1$.

Task 4. Study other exponential kernels. For distinct real nodes $t_0, \ldots, t_{n-1}$, analyze which diagonal entries maximize the inverse of $(e^{st_it_j})$, and identify node conditions that yield a dimension-independent range of maximizing indices.

Task 5. Measure separation from off-diagonal entries. Obtain explicit estimates for the gap between $M_b(n)$ and $\max_{i \ne j} |c_{ij}|$, and determine its behavior for fixed b as n increases.

References

[1] C. Sanna, J. Shallit, and S. Zhang, *The Largest Entry in the Inverse of a Vandermonde Matrix*, arXiv:2008.01012v1 (2020). Accepted manuscript, 11 December 2020: institutional record. Published in *Linear and Multilinear Algebra* **70** (2022), 5634–5641, DOI 10.1080/03081087.2021.1922337.